

Dilip Sahu

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SUMMARY

2nd-year B.Tech CSE student at **NIT Mizoram** seeking **Software Engineering** roles. Specializing in IoT Systems, Scalable Backends, and Applied ML. Proficient in architecting full-stack applications with 3NF-normalized databases, real-time telemetry pipelines, and predictive ML models. Upstream contributor to **OpenCV** and multi-time coding contest winner.

TECHNICAL SKILLS

Languages: C/C++, Java, Python, TypeScript, SQL, Bash
Back-End & Databases: FastAPI, Flask, MySQL (3NF/BCNF), SQLAlchemy, Kafka, RBAC, RESTful APIs
Systems & IoT: ESP32 (C++), GSM/GPRS (SIM800L), UART/Wi-Fi Protocols, Telemetry Failover
AI & ML: Meta Prophet, Random Forest, RAG, ChromaDB, Claude AI, INT8/1-bit Quantization
Developer Tools: Git, Docker, Linux, Leaflet.js (GIS), PM2, Postman
Problem Solving: 400+ Data Structures & Algorithms Problems (LeetCode, Striver A2Z)

EXPERIENCE

OpenCV (Intel-led Open Source) Remote
Sep 2025
Technical Contributor (C++)
– Identified a **floating-point precision regression** in MinGW builds; traced root cause to **default FPU behavior** differences across Windows/Linux platforms. Bug report acknowledged as valid by core maintainers; actively investigating an upstream fix.

VIRTUAL EXPERIENCE

JPMorgan Chase & Co. Forage Simulation
Mar 2026
Software Engineering Virtual Experience
– Implemented a real-time data visualizer using **Apache Kafka** and H2 DB during a virtual engineering simulation; achieved **sub-10ms response latency** for high-frequency financial data ingestion flows.

PROJECTS

IoT Campus Bus Tracker | ESP32, C++, SIM800L, Flask, Leaflet.js Feb 2026 – Apr 2026
– Architected a telemetry system with **dual-network failover** (GSM/Wi-Fi) and 60s timeout, significantly improving data reliability. Implemented an ESP32 **circular buffer (10-packet)** to guarantee data continuity during outages.
– Built a real-time GIS dashboard using **Leaflet.js**, providing sub-second location updates for 1000+ users.

Senior Care Portal | Flask, SQLAlchemy, RBAC, MySQL (3NF) Jan 2026 – May 2026
– Designed a **3NF-compliant 15-table relational schema** with resident lifecycle workflows. Implemented database-level RBAC and **stateless audit logging** for medical data integrity.

TourAI: Smart Travel Planner | FastAPI, Meta Prophet, Claude AI, RAG Mar 2026 – Apr 2026
– Won **1st Place** at ARTHKRITI AI Hackathon. Orchestrated a **6-module ML pipeline** integrating cost forecasting (Meta Prophet) and financial feasibility checks (Random Forest).

Binary Neural Network (BNN) | Java, MNIST Nov 2025 – Jan 2026
– Developed a pure Java neural network with **binary input encoding** and manual backpropagation; attained **99.1% accuracy** on MNIST while optimizing memory footprint.

EDUCATION & LEADERSHIP

National Institute of Technology (NIT), Mizoram Expected May 2028
Aizawl, India
B.Tech in Computer Science and Engineering; CGPA: 7.70/10.00
– **Relevant Coursework:** Data Structures & Algorithms, Database Management Systems (DBMS), Object-Oriented Programming, Operating Systems.
– **Leadership:** Team Head, Coding & AI Club; **Organizer & Problem Setter** for "Trishul" coding contest during TechnoX'25-26; mentoring 15+ students weekly on advanced DSA.
– **Awards:** 2nd Position in Ideathon (NIT Mizoram); Top 48% Globally in Kaggle NLP Challenge (Ranked 1288/2700+).